

Homework #1

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Exercise 2.2

- (a) In these types of problems, we want to solve what $u(x)$ is given $u''(x) = f(x)$ and some additional information. If we set $G''(x; \bar{x}) = \delta(x - \bar{x})$ and are given the conditions that $G'(0) = 0$ and $G(1) = 0$ we can set up some linear algebra to solve for $G(x; \bar{x})$.

As the delta function is defined to be 0 when $x \neq \bar{x}$ we get that $G''(x; \bar{x}) = 0$ when $x \neq \bar{x}$. We also know that the integral of a function must be continuous. Thus, a general solution to this D.E. would be

$$G(x; \bar{x}) = \begin{cases} Ax + B, & 0 \leq x < \bar{x}, \\ Cx + D, & \bar{x} < x \leq 1. \end{cases}$$

We now have four conditions that let us solve this, these being that (as we already stated) G must be continuous (ie. $A\bar{x} + B = C\bar{x} + D$), G has the jump condition at $\bar{x}$ (ie. $C - A = 1$), $G'(0) = 0$, and $G(1) = 0$.

$$\begin{aligned} G'(0) = 0 &\implies A = 0 \\ C - A = 1 &\implies C = 1 \\ G(1) = 0 &\implies C(1) + D = 0 \implies D = -1 \\ A\bar{x} + B = C\bar{x} + D &\implies B = \bar{x} - 1 \end{aligned}$$

So

$$G(x; \bar{x}) = \begin{cases} \bar{x} - 1, & 0 \leq x < \bar{x}, \\ x - 1, & \bar{x} < x \leq 1. \end{cases}$$

We now almost have our solution, and would like to say that $u(x) = \int_0^1 f(\bar{x}) G(x; \bar{x}) d\bar{x}$.

Note, this would be because taking two derivatives of both sides would give you

$$\begin{aligned} u''(x) &= \frac{d^2}{dx^2} \int_0^1 f(\bar{x}) G(x; \bar{x}) d\bar{x} \\ &= \int_0^1 f(\bar{x}) G''(x; \bar{x}) d\bar{x} \\ &= \int_0^1 f(\bar{x}) \delta(\bar{x} - x) d\bar{x} \quad \text{note delta is even, so } \delta(x - \bar{x}) = \delta(\bar{x} - x) \\ &= f(x) \end{aligned}$$

The issue, is that this is also a solution, given the defining properties of G_0 and G_1 in the problem. $u(x) = \alpha G_0(x) + \beta G_1(x) + \int_0^1 f(\bar{x}) G(x; \bar{x}) d\bar{x}$.

So we solve two final (easy) D.E.

$$\begin{aligned} G_0''(x) &= 0 \implies G_0(x) = ax + b \\ G_0'(0) &= 1 \implies a = 1 \\ G_0(1) &= 0 \implies a + b = 0 \implies b = -1 \end{aligned}$$

So $G_0 = x - 1$

$$\begin{aligned} G_1''(x) &= 0 \implies G_1(x) = cx + d \\ G_1'(0) &= 0 \implies c = 0 \\ G_1(1) &= 1 \implies c + d = 1 \implies d = 1 \end{aligned}$$

So $G_1 = 1$

This gives the final solution

$$u(x) = \alpha(x - 1) + \beta + \int_0^1 f(\bar{x}) G(x; \bar{x}) d\bar{x}$$

(b) Figure 2.54 is as below

$$\frac{1}{h^2} \begin{bmatrix} -h & h & 0 & 0 & \cdots & 0 & 0 \\ 1 & -2 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & -2 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 & -2 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 \end{bmatrix} \begin{bmatrix} U_0 \\ U_1 \\ U_2 \\ \vdots \\ U_m \\ U_{m+1} \end{bmatrix} = \begin{bmatrix} \sigma \\ f(x_1) \\ f(x_2) \\ \vdots \\ f(x_m) \\ \beta \end{bmatrix}$$

Consider this as $AU = b$ where the $\frac{1}{h^2}$ is distributed to the matrix A .

Now one at a time, we solve $Av^{(j)} = e_j$ where $v^{(j)}$ is the j th column of the inverse matrix, and e_j is the standard basis vector with a 1 in the j th slot.

For $j = 1, \dots, m$, we get that

$$\frac{1}{h^2}(v_{i-1}^{(j)} - 2v_i^{(j)} + v_{i+1}^{(j)}) = \begin{cases} 1, & j = i, \\ 0, & j \neq i. \end{cases}$$

So

$$v_{i-1}^{(j)} - 2v_i^{(j)} + v_{i+1}^{(j)} = \begin{cases} h^2, & j = i, \\ 0, & j \neq i. \end{cases}$$

This looks a lot like part a, so we'll proceed similarly.

On $j \neq i$, we get that $v_{i-1}^{(j)} - 2v_i^{(j)} + v_{i+1}^{(j)} = 0 \implies v_{i-1}^{(j)} - v_i^{(j)} = v_i^{(j)} - v_{i+1}^{(j)}$

So $v^{(j)}$ has constant slope over the values $i-1, i, i+1$, so we can write $v_i^{(j)} = ai + b$.

So

$$v_i^{(j)} = \begin{cases} a_-i + b_-, & i < j, \\ a_+i + b_+, & i > j. \end{cases}$$

Then we impose some constraints to solve this:

$v_i^{(j)}$ should be well defined, so $a_-j + b_- = a_+j + b_+$

When $i = j$ $v_{j-1}^{(j)} - 2v_j^{(j)} + v_{j+1}^{(j)} = h^2$, so $(a_-(j-1) + b_-) + 2(a_-j + b_-) + (a_+(j+1) + b_+) = h^2$

So $a_+ - a_- = h^2$ (Note that $a_-j + b_- = a_+j + b_+$)

The first and last rows of A also give us information, that being that

$\frac{-v_0^{(j)} + v_1^{(j)}}{h} = 0$ and $v_{m+1}^{(j)} = 0$ respectively.

So $\frac{-b_- + (a_- + b_-)}{h} = \frac{a_-}{h} = 0 \implies a_- = 0$

And $a_+(m+1) + b_+ = 0$

All together this is four constraints which tell us:

$$a_- = 0 \quad (\text{third one})$$

$$a_+ - a_- = h^2 \implies a_+ = h^2 \quad (\text{second one})$$

$$a_+(m+1) + b_+ = 0 \implies -h^2(m+1) = b_+$$

$$a_-j + b_- = a_+j + b_+ \implies b_- = h^2j - h^2(m+1) = h^2(j - m - 1)$$

So for an $m+2 \times m+2$ matrix where $(j, i) \in \{1, \dots, m\} \times \{0, 1, \dots, m, m+1\}$

$$v_i^{(j)} = \begin{cases} h^2(j - m - 1), & i \leq j, \\ h^2i - h^2(m+1), & i > j. \end{cases} = \begin{cases} h^2(j - m - 1), & i \leq j, \\ h^2(i - m - 1), & i > j. \end{cases}$$

Similar calculations for the cases where $j = 0, m+1$ give that $v_i^0 = ih - 1$ and $v_i^{m+1} = 1$

Thus, a 5×5 matrix with $h = 0.25$ would look like

$$A = \begin{bmatrix} -4 & 4 & 0 & 0 & 0 \\ 16 & -32 & 16 & 0 & 0 \\ 0 & 16 & -32 & 16 & 0 \\ 0 & 0 & 16 & -32 & 16 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -1 & -\frac{3}{16} & -\frac{1}{8} & -\frac{1}{16} & 1 \\ -\frac{3}{4} & -\frac{3}{16} & -\frac{1}{8} & -\frac{1}{16} & 1 \\ -\frac{1}{2} & -\frac{1}{8} & -\frac{1}{8} & -\frac{1}{16} & 1 \\ -\frac{1}{4} & -\frac{1}{16} & -\frac{1}{16} & -\frac{1}{16} & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Exercise 2.3

The matrix referenced has a factor of $\frac{1}{h^2}$ which we will omit as it doesn't affect the nullspace. Consider

$$M^T = \begin{bmatrix} -h & 1 & 0 & 0 & \cdots & 0 & 0 \\ h & -2 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & -2 & 1 & \ddots & 0 & 0 \\ 0 & 0 & 1 & -2 & \ddots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & -2 & h \\ 0 & 0 & 0 & 0 & 0 & 1 & -h \end{bmatrix}$$

$$\text{rref}(M^T) = \begin{bmatrix} 1 & 0 & 0 & 0 & \cdots & 0 & -1 \\ 0 & 1 & 0 & 0 & \cdots & 0 & -h \\ 0 & 0 & 1 & 0 & \cdots & 0 & -h \\ 0 & 0 & 0 & 1 & \ddots & 0 & -h \\ \vdots & \vdots & \vdots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 & -h \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 \end{bmatrix}$$

So $\ker(M^T) = \text{span}\{(1, h, \dots, h, 1)\}$.

To verify that $\sigma_1 - \sigma_0 = \frac{h}{2}f(x_0) + h \sum_{i=1}^m f(x_i) + \frac{h}{2}f(x_{m+1})$, just multiply by $\frac{1}{h^2}$ on both sides, row reduce this augmented matrix, and read off the condition of the bottom row which will state that $0 = -\sigma_1 + \sigma_0 + \frac{h}{2}f(x_0) + h \sum_{i=1}^m f(x_i) + \frac{h}{2}f(x_{m+1})$.

Exercise 2.4.prelim

$$\begin{aligned} u'' &= e^x \\ u' &= e^x + c_0 \\ u'(3) &= e^3 + c_0 = \sigma \implies c_0 = \sigma - e^3 \\ u' &= e^x + \sigma - e^3 \\ u &= e^x + \sigma x - e^3 x + c_1 \\ u(0) &= 1 + c_1 = \alpha \implies c_1 = \alpha - 1 \\ u(x) &= e^x + x(\sigma - e^3) + \alpha - 1, 0 < x < 3 \end{aligned}$$

Exercise 2.4

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(a) (Determine the max of the modulus of the entries.)

$$\|e\|_\infty = 5$$

(b) (Sum the modulus of the entries.)

$$\|e\|_1 = 2 + |-5| + 3 + 4 = 14$$

- (c) (Square root the sum of the squares of the entries).

$$\|e\|_2 = \sqrt{4 + 25 + 9 + 16} = \sqrt{54}$$

Exercise A.2

- (a) (Modulus each entry, then find max row sum.)

$$\|A\|_\infty = \max\{10, 26, 42, 58\} = 58$$

- (b) Modulus each entry, then find max col sum.)

$$\|A\|_1 = \max\{28, 32, 37, 40\} = 40$$

- (c) (Square root the maximum modulus of the e-vals of $A^T A$. Use MatLab to compute $A^T A$, then find e-vals and compare. Use command `sqrt(max(abs(eig(B))))`.)

$$\|A\|_2 = \sqrt{\rho(A^T A)} \approx \sqrt{1491.7} \approx 38.6227$$

Exercise A.3

- (a) (Max of the modulus of the function. Find critical point, compare to the end points of 0 and 1, remember to absolute value.)

$$\|u\|_\infty = \max\{|u(0)|, |u(\frac{1}{2})|, |u(1)|\} = \max\{0, \frac{1}{4}, 0\} = \frac{1}{4}$$

- (b) (Integrate modulus from 0 to 1. Split up integrand over positive and negative portion.)

$$\|u\|_1 = \int_0^1 |u(x)| dx = - \int_0^1 u(x) dx = -[\frac{x^3}{3} - \frac{x^2}{2}]_0^1 = -[-\frac{1}{6}] = \frac{1}{6}$$

- (c) (Square root the integral of the square of the modulus from 0 to 1.)

$$\|u(x)\|_2 = \sqrt{\int_0^1 |u(x)|^2 dx} = \sqrt{[\frac{x^5}{5} - \frac{x^4}{2} + \frac{x^3}{3}]_0^1} = \sqrt{\frac{1}{30}} \approx 0.1826$$

Exercise A.4

- (a) (Take the max of the modulus of the outputs at each point. In MatLab, use below code.)

```
>> U = @(x) x.^2 - x;

>> xvals = [.2 .4 .6 .8 1.0];
>> max(abs(U(xvals)))
```

$$\|U\|_\infty = .2400$$

- (b) (This is the sum of the modulus of the outputs, multiplied by the distance between each input.)

$$\|U\|_1 = .2 * \sum_{i=1}^N |U_i| = .2 * (.16 + .24 + .24 + .16 + 0) = .16$$

- (c) (Square root the sum of the square of the modulus of each output multiplied by the interval length.)

$$\|U\|_2 = \sqrt{.2 * (.0256 + .0576 + .0576 + .0256 + 0)} \approx .18243$$

Exercise 1.2

- Step 1: Following the Method of Undetermined Coefficients procedure for any value of $\bar{x}$, define your points near $\bar{x}$ to be $\bar{x} - 2h, \bar{x} - h, \bar{x}, \bar{x} + h, \bar{x} + 2h$ for some distance h between each point.

Step 2: We'll assign the coefficients $c_{-2}, c_{-1}, c_0, c_1, c_2$ to the function's values at these points, respectively. Step 3: This will give

$$\begin{aligned} u''(\bar{x}) &\approx c_{-2}u(\bar{x} - 2h) + c_{-1}u(\bar{x} - h) + c_0u(\bar{x}) + c_1u(\bar{x} + h) + c_2u(\bar{x} + 2h) \\ &= c_{-2} \left(u(\bar{x}) - 2hu'(\bar{x}) + \frac{4h^2}{2}u''(\bar{x}) - \frac{8h^3}{3!}u'''(\bar{x}) + O(h^4) \right) \\ &\quad + c_{-1} \left(u(\bar{x}) - hu'(\bar{x}) + \frac{h^2}{2}u''(\bar{x}) - \frac{h^3}{3!}u'''(\bar{x}) + O(h^4) \right) \\ &\quad + c_0(u(\bar{x})) \\ &\quad + c_1 \left(u(\bar{x}) + hu'(\bar{x}) + \frac{h^2}{2}u''(\bar{x}) + \frac{h^3}{3!}u'''(\bar{x}) + O(h^4) \right) \\ &\quad + c_2 \left(u(\bar{x}) + 2hu'(\bar{x}) + \frac{4h^2}{2}u''(\bar{x}) + \frac{8h^3}{3!}u'''(\bar{x}) + O(h^4) \right) \\ &= u(\bar{x})(c_{-2} + c_{-1} + c_0 + c_1 + c_2) \\ &\quad + u'(\bar{x})(-2hc_{-2} - hc_{-1} + hc_1 + 2hc_2) \\ &\quad + u''(\bar{x}) \left(2h^2c_{-2} + \frac{h^2}{2}c_{-1} + \frac{h^2}{2}c_1 + 2h^2c_2 \right) \\ &\quad + u'''(\bar{x}) \left(-\frac{4h^3}{3}c_{-2} - \frac{h^3}{6}c_{-1} + \frac{h^3}{6}c_1 + \frac{4h^3}{3}c_2 \right) \\ &\quad + u''''(\bar{x}) \left(\frac{2h^4}{3}c_{-2} + \frac{h^4}{24}c_{-1} + \frac{h^4}{24}c_1 + \frac{2h^4}{3}c_2 \right) + \dots \end{aligned}$$

Step 4: Given a large enough expansion of our function at these given terms, we get a 5×5 system:

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ -2 & -1 & 0 & 1 & 2 \\ 2 & \frac{1}{2} & 0 & \frac{1}{2} & 2 \\ -\frac{4}{3} & -\frac{1}{6} & 0 & \frac{1}{6} & \frac{4}{3} \\ \frac{2}{3} & \frac{1}{24} & 0 & \frac{1}{24} & \frac{2}{3} \end{bmatrix} \begin{bmatrix} c_{-2} \\ c_{-1} \\ c_0 \\ c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{h^2} \\ 0 \\ 0 \end{bmatrix}$$

2. (Use `fdstencil` in the way below, the inputs are `fdstencil(derivative order, [first step size, second step size, ...])`)

```
>> fdstencil(2, [-2 -1 0 1 2])
```

The derivative $u'(2)$ of u at x_0 is approximated by

$$\frac{1}{h^2} * [\begin{aligned} &-8.333333333333333e-02 * u(x_0-2*h) + \\ &1.333333333333333e+00 * u(x_0-1*h) + \\ &-2.500000000000000e+00 * u(x_0) + \\ &1.333333333333333e+00 * u(x_0+1*h) + \\ &-8.333333333333333e-02 * u(x_0+2*h) \end{aligned}]$$

For smooth u ,

$$\text{Error} = 0 * h^3 u^{(5)} + -0.01111111 * h^4 u^{(6)} + \dots$$

(This holds the c_i coefficient values and tells us the local truncation error of $O(h^4)$ aka it's fourth order accurate.) To check if the coefficients satisfy our matrix, realize their values as fractions and use this code:

```
>> c = [-1/12; 4/3; -5/2; 4/3; -1/12];
>> M = [ 1      1      1      1      1;
        -2     -1      0      1      2;
         2      1/2    0      1/2    2;
        -4/3   -1/6    0      1/6    4/3;
         2/3    1/24   0      1/24   2/3 ];
>> M*c
```

ans =

```
-0.0000
 0.0000
 1.0000
  0
  0
```

This specifically uses an h value of 1, so there's a secret little $1/h^2$ multiplied with the c vector that we don't see as $h = 1$. In conclusion,

$$u''(\bar{x}) \approx \frac{1}{h^2} \left[-\frac{1}{12}u(\bar{x} - 2h) + \frac{4}{3}u(\bar{x} - h) - \frac{5}{2}u(\bar{x}) + \frac{4}{3}u(\bar{x} + h) - \frac{1}{12}u(\bar{x} + 2h) \right]$$

3. With $u(x) = \sin(2x)$ we get

$$u''(x) \approx \frac{1}{h^2} \left[-\frac{1}{12}\sin(2(1 - 2h)) + \frac{4}{3}\sin(2(1 - h)) - \frac{5}{2}\sin(2(1)) + \frac{4}{3}\sin(2(1 + h)) - \frac{1}{12}\sin(2(1 + 2h)) \right]$$

Following the below code will provide the result:

```

>> t = @(h) (1./h.^2) .* ((-1/12).*sin(2*(1 - 2*h)) + ( 4/3 ).*sin(2*(1 - h)) + (-5/2 ).
>> u2true = -4*sin(2);
hvals = logspace(-1,-4,13);
Et = [];

disp(' ')
disp('          h          Dt (approx u''''(1))          Error')

for i = 1:length(hvals)
    h = hvals(i);
    Dt = t(h);
    Et(i) = Dt - u2true;
    disp(sprintf('%13.4e    % .16e    % .3e', h, Dt, Et(i)))
end

          h          Dt (approx u''(1))          Error
1.0000e-01    -3.6371252766545497e+00    6.443e-05
5.6234e-02    -3.6371832484862447e+00    6.459e-06
3.1623e-02    -3.6371890609225308e+00    6.464e-07
1.7783e-02    -3.6371896426498420e+00    6.465e-08
1.0000e-02    -3.6371897008419252e+00    6.461e-09
5.6234e-03    -3.6371897066651950e+00    6.375e-10
3.1623e-03    -3.6371897072506871e+00    5.204e-11
1.7783e-03    -3.6371897073419088e+00    -3.918e-11
1.0000e-03    -3.6371897078057991e+00    -5.031e-10
5.6234e-04    -3.6371897077851507e+00    -4.824e-10
3.1623e-04    -3.6371897116360685e+00    -4.333e-09
1.7783e-04    -3.6371897172205157e+00    -9.918e-09
1.0000e-04    -3.6371897410569787e+00    -3.375e-08

>> clf
loglog(hvals, abs(Et), 'o-')
grid on
xlabel('h')
ylabel('|error|')
title('Error of t(h) approximating u''''(1)')

```

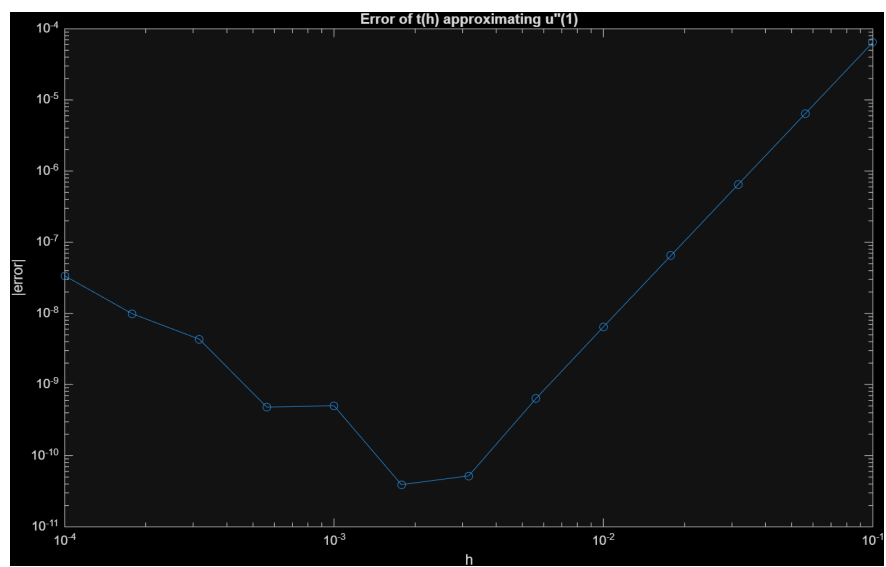


Figure 1: Error vs h for the finite-difference approximation of $u''(1)$.